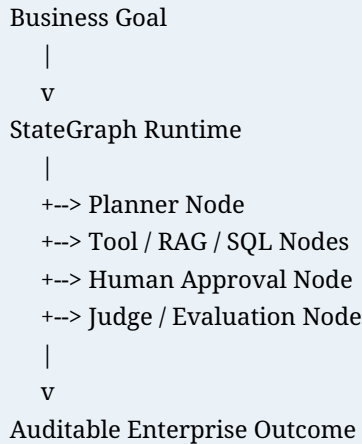


LangGraph Technical Masterclass

End-to-End Guide: From Simple Graphs to Production Multi-Agent Systems



Who this PDF is for

- Layman / business reader: understand why LangGraph matters for real enterprise agents.
- Developer: learn StateGraph, nodes, edges, routers, tools, checkpointing, interrupts, and testing.
- Architect: design secure, observable, scalable, stateful multi-agent systems.
- Interview / whiteboarding: explain LangGraph clearly in 15, 30, and 60 minute formats.

How to read this masterclass

1. Read Chapters 1-5 for the mental model.
2. Use Chapters 6-14 to build real LangGraph workflows.
3. Use Chapters 15-21 for production and enterprise architecture.
4. Use the appendices for interview answers, checklists, and reference code.

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Chapter 1: Why LangGraph Exists

Most simple LLM demos are linear: prompt in, answer out. Enterprise agents are not linear. They need to call tools, retry failed steps, route to specialized skills, ask humans for approval, resume after failures, and leave an audit trail. LangGraph exists to make those stateful, long-running, controllable workflows explicit.

Layman explanation

Imagine a customer support employee handling a complex issue. They read the ticket, check customer data, search policy documents, ask a manager for approval, update the ticket, and reply. A chatbot can answer one question. A LangGraph workflow can model the entire process as steps with memory and control.

Simple Chatbot

User -> LLM -> Answer

Enterprise Agent

User -> Planner -> Tools -> Review -> Action -> Audit -> Answer

Concept	Practical Meaning
Chatbot	Usually a single request-response interaction.
Agent	Can plan, call tools, maintain state, and complete a task.
LangGraph	A graph runtime for controlling how agents move through steps.
Enterprise value	Durability, human review, observability, and governance.

Why normal agent loops become risky

- They can loop indefinitely if the stopping condition is weak.
- They can call risky tools without approval.
- They can lose context after a crash.
- They can be hard to debug because reasoning and execution are mixed.
- They can be difficult to test because every run follows a slightly different path.

LangGraph makes the flow explicit: each node has a responsibility, each edge represents a possible transition, and the state is the shared contract across the system.

When to use LangGraph

- Use it for multi-step workflows, long-running jobs, approval flows, multi-agent orchestration, tool-heavy agents, and workflows that must resume safely.
- Use a simple LLM call or basic chain for small, stateless tasks.
- Use prebuilt agents when you need speed and the risk is low; use LangGraph when you need control.

Low complexity -> Prompt / Chain

Medium -> Tool-calling agent

High -> LangGraph stateful workflow

Enterprise -> LangGraph + checkpointing + HITL + observability

Chapter 2: The Mental Model

LangGraph is easiest to understand as a state machine for AI workflows. The graph holds state. Nodes read state and return updates. Edges decide which node runs next. Conditional edges and commands let the graph route dynamically.

Four core ideas

Concept	Practical Meaning
State	The shared memory object passed through the graph.
Node	A Python function that does one unit of work.
Edge	A connection that decides what runs after a node.
Checkpoint	A saved state snapshot that enables resume and debugging.

```
START
|
v
Node A --updates--> State
|
v
Node B --updates--> State
|
v
END
```

LangGraph vs traditional workflow engines

A traditional workflow engine is good at deterministic business processes. LangGraph is useful when some steps require LLM reasoning, tool selection, uncertain branching, or human review. In production, use both ideas: deterministic rules for critical logic and LLM nodes for language understanding, summarization, planning, and flexible decision support.

- Use code for rules: eligibility thresholds, policy limits, SLAs, risk cutoffs.
- Use LLMs for interpretation: free text, document understanding, summaries, classification, planning suggestions.
- Use judge nodes to validate outputs before taking action.

What a good graph feels like

- Each node has one responsibility.
- State keys are named like business objects, not random variables.
- Routing decisions are easy to explain on a whiteboard.
- Every tool call is traceable.
- Every risky action has a guardrail or human approval path.

Chapter 3: Environment Setup

This chapter provides a practical setup for local development. The examples are written to be easy to read and adapt. In real projects, pin package versions, configure secrets securely, and use proper logging.

Install packages

```
# Recommended: create a clean virtual environment first
python -m venv .venv
source .venv/bin/activate # Windows: .venv\Scripts\activate

pip install -U langgraph langchain langchain-openai python-dotenv pydantic
```

For Azure OpenAI, use your enterprise-approved SDK and environment variables. Keep API keys out of notebooks and source control.

Suggested project structure

```
langgraph-masterclass/
├── app/
│   ├── graphs/
│   │   ├── loan_graph.py
│   │   └── servicenow_graph.py
│   ├── nodes/
│   │   ├── planner.py
│   │   ├── tools.py
│   │   └── validators.py
│   ├── schemas/
│   │   └── state.py
│   ├── tests/
│   │   ├── test_routing.py
│   │   └── test_nodes.py
│   ├── .env
│   ├── pyproject.toml
│   └── README.md
```

This separation keeps graph wiring, node logic, schemas, and tests clean. It also allows different teams to work in parallel: one team builds tools, another builds prompts, another builds guardrails, and another builds deployment.

Minimal imports used throughout

```
from typing import TypedDict, Literal, Annotated
from langgraph.graph import StateGraph, START, END
from langgraph.types import Command, interrupt
```

Chapter 4: Your First Graph

The first graph should be boring. It should prove you understand the mechanics before adding tools, LLMs, memory, and approval flows.

Hello world graph

```
from typing import TypedDict
from langgraph.graph import StateGraph, START, END

class HelloState(TypedDict):
    user_input: str
    response: str

def greet(state: HelloState) -> dict:
    name = state["user_input"]
    return {"response": f"Hello {name}, welcome to LangGraph."}

builder = StateGraph(HelloState)
builder.add_node("greet", greet)
builder.add_edge(START, "greet")
builder.add_edge("greet", END)

app = builder.compile()
result = app.invoke({"user_input": "Mahendra"})
print(result["response"])
```

This example has one state object, one node, one edge from START, one edge to END, and one invoke call. Every larger workflow is built from the same primitives.

Whiteboard explanation

START -> greet -> END

State before greet:

user_input = "Mahendra"

State after greet:

response = "Hello Mahendra, welcome to LangGraph."

Common mistakes

- Returning the full state unnecessarily instead of only changed keys.
- Putting too much work in one node.
- Using unclear state names such as data1, temp, output2.
- Adding dynamic logic before the simple linear graph works.

Chapter 5: State Design

State is the backbone of LangGraph. If state is poorly designed, the graph becomes hard to test, debug, and evolve. Good state design uses business language and separates user input, intermediate artifacts, decisions, and final outputs.

State as a contract

Think of state as the contract between all nodes. Every node should know which keys it reads and which keys it writes.

```
class LoanState(TypedDict, total=False):
    application_id: str
    customer_id: str
    customer_message: str
    documents: list[dict]
    extracted_fields: dict
    credit_score: int
    fraud_risk_score: float
    compliance_status: str
    approval_decision: str
    errors: list[str]
    audit_events: list[dict]
```

State design rules

- Use explicit business names.
- Separate raw input from derived values.
- Store decisions separately from evidence.
- Capture errors and audit events in state.
- Avoid storing huge documents directly in state; store references or summaries when possible.

Bad vs good state

Concept	Practical Meaning
Bad	state["data"] = everything mixed together.
Good	state["credit_score"], state["document_findings"], state["approval_decision"].
Bad	Node returns unrelated keys that surprise downstream nodes.
Good	Node documents which keys it reads and writes.

Chapter 6: Nodes and Edges

Nodes are units of work. Edges define flow. In production, a node should be small enough to test independently but meaningful enough to represent a business step.

Node anatomy

```
def extract_documents(state: LoanState) -> dict:
    """Read uploaded document references and extract structured fields."""
    docs = state.get("documents", [])
    extracted = {}
    for doc in docs:
        # Replace with OCR or document AI call
        extracted[doc["type"]] = {"status": "processed"}
    return {"extracted_fields": extracted}
```

A node receives state and returns updates. It should not mutate global variables. It should handle its own local errors or return structured error information.

Edge patterns

Linear:

START -> A -> B -> END

Branching:

A -> router -> B or C

Loop:

A -> validate -> retry A or END

Human approval:

A -> approval -> action or END

Testing nodes independently

```
def test_extract_documents():
    state = {"documents": [{"type": "bank_statement"}]}
    result = extract_documents(state)
    assert "extracted_fields" in result
    assert result["extracted_fields"]["bank_statement"]["status"] == "processed"
```

Chapter 7: Conditional Routing

Conditional routing is where LangGraph starts feeling powerful. It allows the workflow to decide the next step based on state, business rules, risk score, or LLM output.

Simple router

```
def route_after_credit_check(state: LoanState) -> str:
    score = state.get("credit_score", 0)
    if score >= 750:
        return "low_risk_approval"
    if score >= 600:
        return "manual_review"
    return "reject_application"

builder.add_conditional_edges(
    "credit_check",
    route_after_credit_check,
    {
        "low_risk_approval": "approve",
        "manual_review": "human_review",
        "reject_application": "reject",
    },
)
```

Routing best practices

- Keep high-risk routing deterministic where possible.
- Log why a route was selected.
- Do not route directly from free-form LLM text; convert to structured output first.
- Use explicit route names that can be shown in traces and audit reports.

Whiteboard view

```
Credit Check
|
v
Route by score
+-- score >= 750 -> Approve
+-- 600-749    -> Human Review
+-- score < 600 -> Reject
```

Chapter 8: Reducers and Parallel State Updates

Some workflows run multiple nodes that update the same state key. Reducers define how updates are merged. This is important for messages, audit events, tool results, and parallel agent outputs.

Why reducers matter

If two nodes both return `audit_events`, should the second overwrite the first, or should the events be appended? Reducers answer this question.

```
import operator
from typing import Annotated, TypedDict

class AuditState(TypedDict):
    audit_events: Annotated[list[dict], operator.add]

# If Node A returns {"audit_events": [a]}
# and Node B returns {"audit_events": [b]}
# the final state becomes [a, b]
```

Common reducer use cases

Concept	Practical Meaning
messages	Append chat messages instead of overwriting.
audit_events	Append all operational events.
tool_results	Collect parallel tool outputs.
errors	Collect multiple failures for later handling.
evidence	Merge findings from document, RAG, and SQL agents.

Architectural advice

Use reducers when parallel branches produce additive information. Avoid reducers when only one authoritative decision should exist, such as `final_approval_decision`. For final decisions, use an explicit judge or decision node to reconcile evidence.

Chapter 9: Tool Calling Patterns

A production agent becomes useful when it can call tools. In LangGraph, tools are usually wrapped as nodes or invoked inside nodes. The graph controls when tool calls happen and what happens after.

Tool node as deterministic business capability

```
def get_customer_profile(customer_id: str) -> dict:
    # Replace with real API / database access
    return {
        "customer_id": customer_id,
        "segment": "premier",
        "country": "UK",
        "kyc_status": "verified",
    }

def customer_profile_node(state: LoanState) -> dict:
    profile = get_customer_profile(state["customer_id"])
    return {"customer_profile": profile}
```

Tool safety levels

Concept	Practical Meaning
Read-only tools	Generally lower risk: search, retrieve, lookup.
Write tools	Higher risk: update CRM, create ticket, send email.
Financial tools	Require approval, limits, audit, and rollback strategy.
Administrative tools	Require strong RBAC and human approval.
External communication	Usually requires review before sending.

Pattern: tool call followed by judge

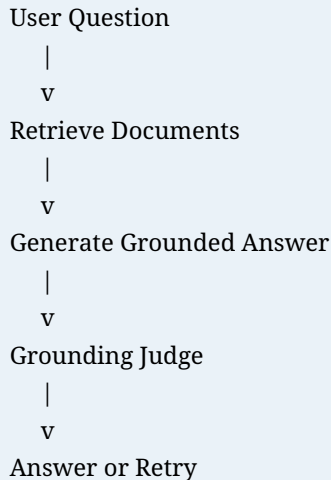
```
Tool Node -> Validate Result -> Continue
    -> If invalid -> Retry / Escalate
```

Never assume a tool result is correct just because the tool returned successfully. Validate schema, freshness, permissions, and business consistency.

Chapter 10: RAG with LangGraph

RAG is retrieval augmented generation. LangGraph does not replace RAG; it orchestrates RAG as one capability within a larger workflow. A RAG agent can retrieve policy evidence, summarize it, and pass evidence to a judge or decision node.

RAG node flow



RAG node pseudo-code

```
def retrieve_policy_node(state: dict) -> dict:
    query = state["question"]
    docs = vector_store.similarity_search(query, k=5)
    evidence = [
        {"source": d.metadata.get("source"), "content": d.page_content}
        for d in docs
    ]
    return {"evidence": evidence}

def answer_with_evidence_node(state: dict) -> dict:
    prompt = f"""
    Answer using only this evidence:
    {state['evidence']}

    Question: {state['question']}
    """
    answer = llm.invoke(prompt)
    return {"draft_answer": answer.content}
```

Enterprise RAG controls

- Metadata filters for tenant, country, product, policy version, and access group.
- Hybrid search when keyword exactness matters.
- Re-ranking for better context precision.
- Citations and source tracking in state.

- Groundedness evaluation before final response.

Chapter 11: SQL and Enterprise Tool Nodes

Many enterprise agents need structured data. SQL nodes are powerful but risky. The safest pattern is to expose controlled, parameterized tools instead of letting the LLM write arbitrary SQL.

Unsafe vs safe SQL

Concept	Practical Meaning
Unsafe	LLM writes arbitrary SQL against production database.
Safer	LLM selects from approved tools with parameterized queries.
Best	Read-only views, RBAC, query limits, audit logs, and human approval for risky actions.

Parameterized SQL tool

```
def get_open_tickets(customer_id: str) -> list[dict]:
    sql = """
    SELECT ticket_id, status, priority, created_at
    FROM support_tickets
    WHERE customer_id = :customer_id
    ORDER BY created_at DESC
    LIMIT 20
    """
    return db.query(sql, {"customer_id": customer_id})

def open_tickets_node(state: dict) -> dict:
    tickets = get_open_tickets(state["customer_id"])
    return {"open_tickets": tickets}
```

Enterprise safeguards

- Use read-only credentials for retrieval agents.
- Use stored procedures or service APIs for writes.
- Apply row-level security and tenant isolation.
- Log every query with user, thread_id, purpose, and result count.
- Block destructive SQL operations from agent execution paths.

Chapter 12: Persistence and Checkpointing

Persistence is one of LangGraph's most important production features. A checkpointer saves graph state at execution steps so the graph can resume, support memory, enable human-in-the-loop, and help with debugging.

Why checkpointing matters

```
Run starts
|
Node A completes -> checkpoint
|
Node B fails
|
Resume from checkpoint instead of starting over
```

- Resume after crash or timeout.
- Pause for human approval and continue later.
- Inspect historical state for debugging.
- Replay or fork from previous checkpoints.
- Maintain conversational memory by thread_id.

In-memory checkpointer for learning

```
from langgraph.checkpoint.memory import MemorySaver

memory = MemorySaver()
app = builder.compile(checkpointer=memory)

config = {"configurable": {"thread_id": "loan-123"}}
app.invoke({"customer_id": "C001"}, config=config)

# Later, use the same thread_id to continue the same thread.
state_snapshot = app.get_state(config)
print(state_snapshot.values)
```

Production advice

Use durable storage for production checkpointing. In-memory checkpointing is fine for local demos but not for enterprise reliability. Store only necessary state, encrypt sensitive fields, and define retention policies for compliance.

Concept	Practical Meaning
Dev	MemorySaver for local learning.
Test	Persistent checkpointer with test database.
Production	Durable checkpointer, encryption, backups, monitoring, retention policy.

Chapter 13: Human-in-the-Loop with Interrupts

Human-in-the-loop is essential when the agent is about to take a risky action or when confidence is low. LangGraph interrupts let a graph pause, surface information to a human, and resume from the same state.

Approval node

```
from langgraph.types import interrupt, Command

class ApprovalState(TypedDict, total=False):
    proposed_action: str
    approved: bool
    human_comment: str

def human_approval_node(state: ApprovalState) -> dict:
    decision = interrupt({
        "question": "Approve this action?",
        "proposed_action": state["proposed_action"],
    })
    return {
        "approved": decision.get("approved", False),
        "human_comment": decision.get("comment", ""),
    }
```

Where to place approval

- Before sending external emails.
- Before updating customer records.
- Before financial decisions or refunds.
- Before closing high-severity ServiceNow incidents.
- When confidence or groundedness score falls below threshold.

Approval flow

```
Draft Action
|
v
Human Approval Interrupt
|
+-- Approved -> Execute Action
+-- Edited   -> Execute Edited Action
+-- Rejected -> Stop / Escalate
```

Chapter 14: Error Handling, Retries, and Idempotency

Agents fail in ways normal applications do not: tools time out, LLMs return malformed output, routing may be wrong, and humans may pause workflows for days. Build for failure from the beginning.

Retry policy pattern

```
from langgraph.types import RetryPolicy

builder.add_node(
    "search_policy",
    search_policy_node,
    retry_policy=RetryPolicy(max_attempts=3),
)
```

Retries should be used for transient failures such as timeouts, rate limits, or temporary service errors. Do not blindly retry deterministic business validation failures.

Idempotency

Idempotency means running the same action twice should not create duplicate side effects. This matters because a graph may resume after a failure and repeat a node unless the action is designed safely.

- Use idempotency keys for ticket creation, payment requests, and emails.
- Check whether an action already happened before repeating it.
- Store external action IDs in state.
- Separate `prepare_action` from `execute_action` so approval and execution are traceable.

Failure handling table

Concept	Practical Meaning
Tool timeout	Retry with backoff, then escalate.
Bad LLM JSON	Re-prompt with schema, then fallback.
Low confidence	Route to human review.
Permission denied	Stop and notify owner; do not bypass RBAC.
Duplicate action risk	Use idempotency key and action log.

Chapter 15: Streaming and User Experience

Long-running agents need good user experience. Streaming lets the application show progress while the graph is still running. For business users, this builds trust because they can see what the agent is doing.

What to stream

- Progress events: reading documents, checking credit, searching policies.
- Intermediate summaries: evidence found, tool results, pending approval.
- Final answer chunks when generating a response.
- Interrupt events when human approval is needed.

Example event model

```
for event in app.stream(input_state, config=config):  
    print(event)
```

```
# In a web app, convert these events into Server-Sent Events or WebSocket messages.
```

UX pattern

Frontend

|

+-- User submits task

+-- Shows progress stream

+-- Shows approval card if interrupted

+-- Shows final result and audit summary

Chapter 16: Memory Patterns

Memory in LangGraph can mean short-term conversation memory, checkpointed workflow state, long-term semantic memory, or business history. Do not mix them blindly.

Memory types

Concept	Practical Meaning
Working memory	Current graph state and messages.
Thread memory	Checkpointed state tied to a <code>thread_id</code> .
Semantic memory	Facts stored and retrieved by embedding search.
Episodic memory	Historical events such as past approvals, incidents, or actions.
Business memory	Authoritative data from CRM, ERP, ServiceNow, or banking systems.

Design guidance

- Do not store regulated data in vector stores without governance.
- Keep authoritative facts in systems of record.
- Use semantic memory for helpful context, not as the source of truth for regulated decisions.
- Make memory writes explicit and auditable.
- Allow users or compliance teams to inspect and delete stored memory where required.

Memory architecture

LangGraph State

|

+-- Checkpointer: resume and thread memory

+-- Vector Store: semantic memory

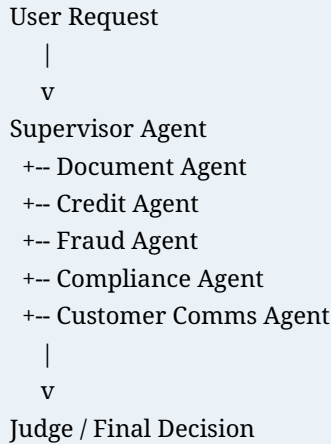
+-- Postgres: episodic audit history

+-- CRM / Core System: source of truth

Chapter 17: Multi-Agent Supervisor Pattern

The supervisor pattern uses one coordinator to route tasks to specialized agents. It is easy to explain, govern, and observe. This is often the best starting point for enterprise multi-agent systems.

Supervisor architecture



Simple supervisor router

```
def supervisor_router(state: dict) -> str:
    task = state.get("current_task", "").lower()
    if "document" in task or "ocr" in task:
        return "document_agent"
    if "credit" in task or "score" in task:
        return "credit_agent"
    if "fraud" in task:
        return "fraud_agent"
    if "policy" in task or "compliance" in task:
        return "compliance_agent"
    return "general_agent"
```

Enterprise advice

- Keep supervisor logic transparent and testable.
- Give each specialist agent a clear contract.
- Use separate prompts and tools per specialist.
- Centralize final decision in a judge or approval node.
- Capture which agent produced which evidence.

Chapter 18: Swarm and Handoff Pattern

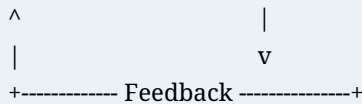
In a swarm pattern, agents hand off control to each other based on specialization. This can be flexible, but it requires strong controls to prevent confusion, loops, and unclear ownership.

Swarm vs supervisor

Concept	Practical Meaning
Supervisor	Central coordinator decides who acts next.
Swarm	Agents can hand off to one another.
Supervisor strength	Governance, auditability, easier testing.
Swarm strength	Flexible collaboration when tasks are ambiguous.
Enterprise recommendation	Start with supervisor; use swarm only where handoff flexibility is truly needed.

Handoff diagram

Research Agent -> Data Agent -> Compliance Agent -> Writer Agent



Controls for swarm systems

- Max handoff count.
- Allowed handoff matrix.
- Clear owner for final response.
- Loop detection.
- Trace all handoff reasons.

Chapter 19: End-to-End Mortgage Origination Blueprint

This chapter turns the LangGraph patterns into a complete enterprise design. The mortgage origination workflow is a strong interview and client workshop example because it includes documents, structured data, risk, compliance, approvals, and customer communication.

Business process

32. Customer submits mortgage application and documents.
33. Document agent extracts income, address, identity, bank statements, and missing fields.
34. Credit agent checks bureau score and affordability.
35. Fraud agent checks device, identity, duplicate application, and blacklist signals.
36. Compliance agent checks policy and regulatory constraints.
37. Approval agent proposes approve, reject, or manual review.
38. Human reviewer handles exceptions and high-risk cases.
39. Customer communication agent drafts a clear response.

Graph architecture

```
START
|
v
intake_application
|
v
extract_documents
|
+--> credit_check
+--> fraud_check
+--> policy_rag_check
|
v
combine_evidence
|
v
approval_router
+--> auto_approve
+--> human_review
+--> reject
|
v
customer_response
|
v
END
```

State schema

```
class MortgageState(TypedDict, total=False):
    application_id: str
```



```
customer_id: str
documents: list[dict]
extracted_income: float
extracted_address: str
missing_documents: list[str]
credit_score: int
affordability_ratio: float
fraud_score: float
policy_findings: list[dict]
evidence_summary: str
decision: Literal["approve", "review", "reject"]
human_notes: str
customer_response: str
audit_events: list[dict]
```

Risk-based routing

```
def approval_router(state: MortgageState) -> str:
    if state.get("missing_documents"):
        return "human_review"
    if state.get("fraud_score", 1.0) > 0.7:
        return "human_review"
    if state.get("credit_score", 0) < 600:
        return "reject"
    if state.get("affordability_ratio", 1.0) <= 0.35:
        return "auto_approve"
    return "human_review"
```

Key lesson

The LLM should help summarize documents and policy evidence. The actual approval thresholds should remain deterministic and auditable. This is the difference between a demo agent and a production-grade enterprise agent.

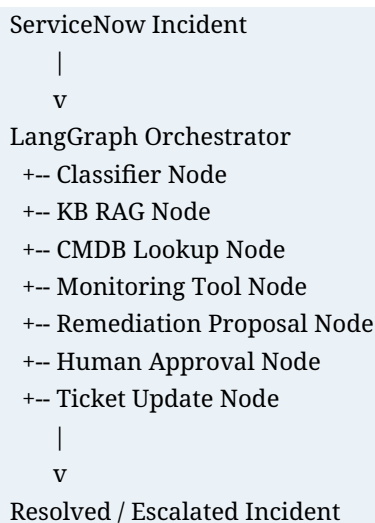
Chapter 20: ServiceNow ITSM Automation Blueprint

ServiceNow is a strong enterprise use case because ticketing workflows already have structured states, SLAs, assignment groups, knowledge articles, and approval rules. LangGraph can orchestrate the reasoning and tool usage around those systems.

Business process

40. New incident arrives.
41. Classifier agent identifies category, urgency, affected CI, and assignment group.
42. RAG agent searches knowledge base and past incidents.
43. Diagnostic agent calls monitoring tools.
44. Remediation agent proposes action.
45. Human approval is required for risky remediation.
46. Agent updates ticket, creates KB draft, and sends user communication.

Architecture diagram



Controlled remediation pattern

```
def remediation_router(state: dict) -> str:
    severity = state.get("severity")
    action = state.get("proposed_action", "")
    if severity in ["P1", "P2"]:
        return "human_approval"
    if "restart" in action.lower():
        return "human_approval"
    return "execute_low_risk_action"
```

Executive value

- Faster triage and assignment.
- Better consistency in root-cause analysis.
- Reduced manual effort for low-risk incidents.

- Improved knowledge base quality.
- Auditable automation with human controls.

Chapter 21: Production Architecture on Azure and AWS

A LangGraph application is not just Python code. Production architecture includes identity, API security, state storage, model access, vector search, observability, CI/CD, and governance.

Azure reference architecture

```
User / Business App
|
v
Azure Front Door / App Gateway
|
v
FastAPI on AKS or Container Apps
|
v
LangGraph Runtime
+-- Azure OpenAI
+-- Azure AI Search / pgVector
+-- Redis Cache
+-- PostgreSQL Checkpointer
+-- Key Vault
+-- ServiceNow / SAP / SQL APIs
|
v
App Insights / Log Analytics / LangSmith or Langfuse
```

AWS reference architecture

```
User / Business App
|
v
CloudFront / API Gateway
|
v
EKS / ECS / Lambda
|
v
LangGraph Runtime
+-- Bedrock Models
+-- OpenSearch / Aurora pgvector
+-- ElastiCache Redis
+-- RDS Checkpointer
+-- Secrets Manager
+-- Enterprise APIs
|
v
CloudWatch / X-Ray / LangSmith or Langfuse
```

Architecture checklist

- Use private networking for model, database, and search services where possible.
- Use managed identity / IAM roles instead of long-lived secrets.
- Separate dev, test, staging, and production graphs.
- Trace every run with correlation IDs and thread IDs.
- Define retention and encryption for checkpoints and memory.

Chapter 22: Testing and Evaluation

Testing LangGraph systems requires testing deterministic nodes, routing, tool contracts, prompts, agent trajectories, and business outcomes. Do not rely only on manual chat testing.

Testing layers

Concept	Practical Meaning
Unit tests	Node functions, routers, reducers.
Contract tests	Tool schemas, API responses, permissions.
Golden-set tests	Known inputs with expected outputs or decisions.
Trajectory tests	Expected path through graph nodes.
Evaluation tests	Groundedness, relevance, hallucination, task success.
Load tests	Concurrency, latency, checkpoint storage, tool rate limits.

Router unit test

```
def test_low_credit_routes_to_reject():
    state = {"credit_score": 580, "fraud_score": 0.1, "missing_documents": []}
    assert approval_router(state) == "reject"

def test_high_fraud_routes_to_review():
    state = {"credit_score": 780, "fraud_score": 0.9, "missing_documents": []}
    assert approval_router(state) == "human_review"
```

Evaluation metrics

- Task success rate: did the workflow complete the business task?
- Routing accuracy: did it choose the correct path?
- Tool success rate: did tools return valid results?
- Groundedness: is the answer supported by retrieved evidence?
- Hallucination rate: did it invent facts?
- Human override rate: how often did humans correct the agent?
- Cost per completed task and latency per node.

Chapter 23: Security, Governance, and Auditability

Enterprise LangGraph systems must be secure by design. The graph should enforce permissions, limit tools, protect sensitive data, record decisions, and support audit review.

Security principles

- Least privilege for every tool.
- No direct production database writes from LLM-generated code.
- PII masking where not needed.
- Tenant-aware retrieval and row-level security.
- Human approval for high-impact actions.
- Immutable audit trail for decisions and tool calls.

Governed execution pattern

```
graph TD;
    A[User Request] --> B[AuthN / AuthZ];
    B --> C[Policy Filter];
    C --> D[Allowed Graph Nodes];
    D --> E[Allowed Tools];
    E --> F[Audit Log];
```

Audit event example

```
def audit(state: dict, event_type: str, details: dict) -> dict:
    event = {
        "thread_id": state.get("thread_id"),
        "application_id": state.get("application_id"),
        "event_type": event_type,
        "details": details,
    }
    return {"audit_events": [event]}
```

Chapter 24: Observability and Tracing Patterns

If you cannot see what the agent did, you cannot trust it. Observability for LangGraph should capture graph path, state transitions, tool calls, prompts, model responses, latency, cost, errors, and human decisions.

What to trace

- Input and output for each node, with sensitive fields redacted.
- Routing decisions and reasons.
- Tool name, arguments, result status, latency, and errors.
- Model name, token usage, latency, and cost.
- Checkpoint ID and thread ID.
- Human approval decisions and comments.

Operational dashboard

Metrics

- +-- Completed runs
- +-- Failed runs
- +-- Average latency
- +-- Cost per task
- +-- Tool failure rate
- +-- Human override rate
- +-- Hallucination / groundedness scores

Production incident questions

- Which node failed?
- What state entered that node?
- Which tool was called?
- Was the tool result valid?
- Did routing follow expected logic?
- Can we replay from a checkpoint?
- Was human approval required and captured?

Chapter 25: Deployment and CI/CD

Deployment is where many agent demos fail. Treat the graph like production software: version it, test it, deploy it safely, monitor it, and roll it back when needed.

CI/CD pipeline

```
Commit
|
v
Lint + Unit Tests
|
v
Prompt / Golden-set Evaluation
|
v
Build Container
|
v
Deploy to Dev
|
v
Integration Tests
|
v
Deploy to Staging
|
v
Canary Production
|
v
Full Production
```

Versioning strategy

- Version graph code.
- Version prompts.
- Version tool schemas.
- Version evaluation datasets.
- Version model configuration.
- Capture graph version in every audit event.

Rollback

Rollback is not only code rollback. You may need to roll back prompts, model version, routing thresholds, tool schema, or vector index version. Every run should record enough metadata to explain which configuration produced the result.

Chapter 26: Whiteboarding Playbook

This chapter gives you ready-to-use explanations for interviews, client workshops, and executive discussions.

15-minute version

47. Define LangGraph as a stateful orchestration framework for agents.
48. Draw START -> Planner -> Tool -> Validator -> END.
49. Explain state, nodes, edges, checkpointing.
50. Add human approval for risky actions.
51. Close with production concerns: security, tracing, evaluation.

30-minute version

52. Start with chatbot vs agent vs LangGraph.
53. Explain StateGraph primitives.
54. Build a simple loan flow.
55. Add RAG, SQL, and tool nodes.
56. Add checkpointing and human-in-the-loop.
57. Add supervisor multi-agent architecture.
58. Discuss production deployment and governance.

One-hour client workshop

59. Business process discovery.
60. Agent opportunity mapping.
61. Tool and data source inventory.
62. Risk classification and approval points.
63. LangGraph architecture sketch.
64. Evaluation and observability design.
65. Implementation roadmap and ROI model.

Core diagram to draw

```
User
|
v
API / UI
|
v
LangGraph Orchestrator
+-- Planner
+-- RAG Agent
+-- SQL Agent
+-- Tool Agent
+-- Compliance / Judge
+-- Human Approval
|
v
```


Chapter 27: Capstone Project Code Walkthrough

This capstone combines the masterclass concepts into a simplified but realistic graph. It is intentionally compact so you can explain it in an interview or use it as a starting template.

Capstone state

```
from typing import TypedDict, Literal, Annotated
import operator

class CapstoneState(TypedDict, total=False):
    thread_id: str
    user_request: str
    customer_id: str
    intent: Literal["loan", "ticket", "policy", "unknown"]
    evidence: Annotated[list[dict], operator.add]
    tool_results: Annotated[list[dict], operator.add]
    draft_answer: str
    risk_level: Literal["low", "medium", "high"]
    approved: bool
    final_answer: str
    audit_events: Annotated[list[dict], operator.add]
```

Core nodes

```
def classify_intent(state: CapstoneState) -> dict:
    text = state["user_request"].lower()
    if "loan" in text or "mortgage" in text:
        intent = "loan"
    elif "ticket" in text or "incident" in text:
        intent = "ticket"
    elif "policy" in text:
        intent = "policy"
    else:
        intent = "unknown"
    return {"intent": intent, "audit_events": [{"event": "intent", "value": intent}]}

def retrieve_evidence(state: CapstoneState) -> dict:
    return {"evidence": [{"source": "policy", "content": "Mock policy evidence"}]}

def call_business_tool(state: CapstoneState) -> dict:
    return {"tool_results": [{"tool": "customer_lookup", "status": "ok"}]}

def assess_risk(state: CapstoneState) -> dict:
    if state.get("intent") == "loan":
        return {"risk_level": "high"}
    return {"risk_level": "low"}

def draft_response(state: CapstoneState) -> dict:
    return {"draft_answer": f"Intent: {state['intent']}. Evidence and tools checked."}
```

Human approval and finalization

```
from langgraph.types import interrupt

def approval_if_needed(state: CapstoneState) -> dict:
    if state.get("risk_level") == "high":
        decision = interrupt({
            "question": "Approve high-risk response?",
            "draft_answer": state.get("draft_answer"),
        })
        return {"approved": decision.get("approved", False)}
    return {"approved": True}

def finalize(state: CapstoneState) -> dict:
    if not state.get("approved"):
        return {"final_answer": "This request has been escalated to a human reviewer."}
    return {"final_answer": state.get("draft_answer", "Completed.")}
```

Graph wiring

```
from langgraph.graph import StateGraph, START, END
from langgraph.checkpoint.memory import MemorySaver

builder = StateGraph(CapstoneState)
builder.add_node("classify_intent", classify_intent)
builder.add_node("retrieve_evidence", retrieve_evidence)
builder.add_node("call_business_tool", call_business_tool)
builder.add_node("assess_risk", assess_risk)
builder.add_node("draft_response", draft_response)
builder.add_node("approval_if_needed", approval_if_needed)
builder.add_node("finalize", finalize)

builder.add_edge(START, "classify_intent")
builder.add_edge("classify_intent", "retrieve_evidence")
builder.add_edge("retrieve_evidence", "call_business_tool")
builder.add_edge("call_business_tool", "assess_risk")
builder.add_edge("assess_risk", "draft_response")
builder.add_edge("draft_response", "approval_if_needed")
builder.add_edge("approval_if_needed", "finalize")
builder.add_edge("finalize", END)

app = builder.compile(checkpointer=MemorySaver())
config = {"configurable": {"thread_id": "capstone-001"}}
result = app.invoke({"user_request": "Check my mortgage loan status"}, config=config)
print(result)
```

How to extend the capstone

- Replace mock retrieval with Azure AI Search, pgVector, OpenSearch, or another vector database.
- Replace mock tool calls with MCP, ServiceNow, Salesforce, SAP, or SQL APIs.
- Add structured output validation with Pydantic.

- Add LangSmith or Langfuse tracing.
- Add durable checkpointing using production storage.
- Add golden-set tests for routing and final answers.

Appendix A: Interview Questions and Short Answers

- What is LangGraph?

LangGraph is a stateful orchestration framework/runtime for building long-running, controllable agent workflows.

- Why not just use a normal LangChain agent?

A simple agent is fine for basic tasks; LangGraph is better when you need durable state, routing, human review, checkpointing, multi-agent orchestration, and production control.

- What is StateGraph?

A graph where nodes operate on shared state and edges control execution flow.

- What is a node?

A function that reads state and returns state updates.

- What is an edge?

A connection that defines what runs next.

- What is checkpointing?

Saving graph state at steps so the workflow can resume, support memory, and enable debugging.

- What is interrupt?

A way to pause graph execution, surface a payload to a caller or human, and resume later with input.

- How do you prevent loops?

Use max iterations, clear stopping conditions, route guards, loop counters in state, and tests for cyclic paths.

- Where do you keep business rules?

Deterministic business rules should be code nodes or tools; LLMs should support interpretation, summarization, and reasoning where flexibility is needed.

- How do you secure tools?

Least privilege, read-only defaults, RBAC, parameterized APIs, audit logs, human approval for writes, and blocked destructive operations.

Appendix B: Production Readiness Checklist

- Graph state schema reviewed and documented.
- Every node has unit tests.
- Routers have deterministic test cases.
- Tool contracts are versioned.
- Prompts are versioned and reviewed.
- Checkpointing configured with durable storage.
- PII and secrets are protected.
- Human approval paths exist for risky actions.
- Observability captures node path, tool calls, latency, cost, and errors.
- Golden-set evaluation runs in CI/CD.
- Rollback strategy covers code, prompts, tools, and model configuration.
- Audit events are immutable and searchable.
- Runbooks exist for tool outage, model outage, high hallucination rate, and stuck interrupts.

Appendix C: Glossary

Term	Meaning
Agent	An AI system that can reason, choose actions, use tools, and pursue a goal.
State	The shared data object that flows through a LangGraph workflow.
Node	A function or runnable step in a graph.
Edge	A transition from one node to another.
Conditional edge	A route chosen based on state or logic.
Reducer	A merge strategy for state updates.
Checkpoint	A saved snapshot of state.
Thread ID	Identifier used to continue a checkpointed conversation or workflow.
Interrupt	A pause point for external or human input.
Supervisor	A coordinating agent that routes work to specialized agents.
Handoff	Transfer of control from one agent to another.
RAG	Retrieval augmented generation, where documents are retrieved before generation.
MCP	Model Context Protocol, a standard way of exposing tools and context to models/agents.

References and Further Reading

- LangGraph overview - official LangChain docs: <https://docs.langchain.com/oss/python/langgraph/overview>
- LangGraph persistence - official LangChain docs: <https://docs.langchain.com/oss/python/langgraph/persistence>
- LangGraph interrupts - official LangChain docs: <https://docs.langchain.com/oss/python/langgraph/interrupts>
- Thinking in LangGraph - official LangChain docs: <https://docs.langchain.com/oss/python/langgraph/thinking-in-langgraph>
- LangGraph GitHub repository: <https://github.com/langchain-ai/langgraph>
- LangGraph swarm reference: <https://reference.langchain.com/python/langgraph-swarm>
- LangSmith documentation for tracing and evaluation: <https://docs.smith.langchain.com/>
- LangChain documentation for models, tools, and agents: <https://docs.langchain.com/>

Source note

This handbook combines official LangGraph documentation concepts with practical enterprise architecture patterns, simplified examples, and implementation guidance. Code is illustrative and should be adapted, secured, tested, and version-pinned before production use.